

The glucose in the blood is called blood sugar. Blood sugar levels are regulated by glucose-binding nerve cells in the hypothalamus.^[77] In addition, glucose in the brain binds to glucose receptors of the reward system in the nucleus accumbens.^[77] The binding of glucose to the sweet receptor on the tongue induces a release of various hormones of energy metabolism, either through glucose or through other sugars, leading to an increased cellular uptake and lower blood sugar levels.^[78] Artificial sweeteners do not lower blood sugar levels.^[78]

The blood sugar content of a healthy person in the short-time fasting state, e.g. after overnight fasting, is about 70 to 100 mg/dL of blood (4 to 5.5 mM). In blood plasma, the measured values are about 10-15 % higher. In addition, the values in the arterial blood are higher than the concentrations in the venous blood since glucose is absorbed into the tissue during the passage of the capillary bed. Also in the capillary blood, which is often used for blood sugar determination, the values are sometimes higher than in the venous blood. The glucose content of the blood is regulated by the hormones insulin, incretin and glucagon.^{[77][79]} Insulin lowers the glucose level, glucagon increases it.^[35] Furthermore, the hormones adrenaline, thyroxine, glucocorticoids, somatotropin and adrenocorticotropin lead to an increase in the glucose level.^[35] There is also a hormone-independent regulation, which is referred to as glucose autoregulation.^[80] After food intake the blood sugar concentration increases. Values over 180 mg/dL in venous whole blood are pathological and are termed hyperglycemia, values below 40 mg/dL are termed hypoglycaemia.^[81] When needed, glucose is released into the bloodstream by glucose-6-phosphatase from glucose-6-phosphate originating from liver and kidney glycogen, thereby regulating the homeostasis of blood glucose concentration.^{[53][34]} In ruminants, the blood glucose concentration is lower (60 mg/dL in cattle and 40 mg/dL in sheep), because the carbohydrates are converted more by their gut flora into short-chain fatty acids.^[82]

Some glucose is converted to lactic acid by astrocytes, which is then utilized as an energy source by brain cells; some glucose is used by intestinal cells and red blood cells, while the rest reaches the liver, adipose tissue and muscle cells, where it is absorbed and stored as glycogen (under the influence of insulin). Liver cell glycogen can be converted to glucose and returned to the blood when insulin is low or absent; muscle cell glycogen is not returned to the blood because of a lack of enzymes. In fat cells, glucose is used to power reactions that synthesize some fat types and have other purposes. Glycogen is the body's "glucose energy storage" mechanism, because it is much more "space efficient" and less reactive than glucose itself.

As a result of its importance in human health, glucose is an analyte in glucose tests that are common medical blood tests.^[83] Eating or fasting prior to taking a blood sample has an effect on analyses for glucose in the blood; a high fasting glucose blood sugar level may be a sign of prediabetes or diabetes mellitus.^[84]

The glycemic index is an indicator of the speed of resorption and conversion to blood glucose levels from ingested carbohydrates, measured as the area under the curve of blood glucose levels after consumption in comparison to glucose (glucose is defined as 100).^[85] The clinical importance of the glycemic index is controversial,^{[85][86]} as foods with high fat contents slow the resorption of carbohydrates and lower the glycemic index, e.g. ice cream.^[86] An alternative indicator is the insulin index,^[87] measured as the impact of carbohydrate consumption on the blood insulin levels. The glycemic load is an indicator for the amount of glucose added to blood glucose levels after consumption, based on the glycemic index and the amount of consumed food.

Precursor

Organisms use glucose as a precursor for the synthesis of several important substances. Starch, cellulose, and glycogen ("animal starch") are common glucose polymers (polysaccharides). Some of these polymers (starch or glycogen) serve as energy stores, while others (cellulose and chitin, which is made from a derivative of glucose) have structural roles. Oligosaccharides of glucose combined with other sugars serve as important energy stores. These include lactose, the predominant sugar in milk, which is a glucose-galactose disaccharide, and sucrose, another disaccharide which is composed of glucose and fructose. Glucose is also added onto certain proteins and lipids in a process called glycosylation. This is often critical for their functioning. The enzymes that join glucose to other molecules usually use phosphorylated glucose to power the formation of the new bond by coupling it with the breaking of the glucose-phosphate bond.

Other than its direct use as a monomer, glucose can be broken down to synthesize a wide variety of other biomolecules. This is important, as glucose serves both as a primary store of energy and as a source of organic carbon. Glucose can be broken down and converted into lipids. It is also a precursor for the synthesis of other important molecules such as vitamin C (ascorbic acid). In living organisms, glucose is converted to several other chemical compounds that are the starting material for various metabolic pathways. Among them, all other monosaccharides^[88] such as fructose (via the polyol pathway),^[42] mannose (the epimer of glucose at position 2), galactose (the epimer at position 4), fucose, various uronic acids and the amino sugars are produced from glucose.^[44] In addition to the phosphorylation to glucose-6-phosphate, which is part of the glycolysis, glucose can be oxidized during its degradation to glucono-1,5-lactone. Glucose is used in some bacteria as a building block in the trehalose or the dextran biosynthesis and in animals as a building block of glycogen. Glucose can also be converted from bacterial xylose isomerase to fructose. In addition, glucose metabolites produce all nonessential amino acids, sugar alcohols such as mannitol and sorbitol, fatty acids, cholesterol and nucleic acids.^[88] Finally, glucose is used as a building block in the glycosylation of proteins to glycoproteins, glycolipids, peptidoglycans, glycosides and other substances (catalyzed by glycosyltransferases) and can be cleaved from them by glycosidases.

Pathology

Diabetes

Diabetes is a metabolic disorder where the body is unable to regulate levels of glucose in the blood either because of a lack of insulin in the body or the failure, by cells in the body, to respond properly to insulin. Each of these situations can be caused by persistently high elevations of blood glucose levels, through pancreatic burnout and insulin resistance. The pancreas is the organ responsible for the secretion of the hormones insulin and glucagon.^[89] Insulin is a hormone that regulates glucose levels,

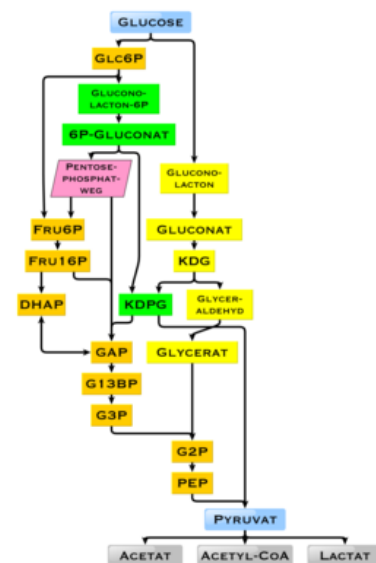


Diagram showing the possible intermediates in glucose degradation; Metabolic pathways orange: glycolysis, green: Entner-Doudoroff pathway, phosphorylating, yellow: Entner-Doudoroff pathway, non-phosphorylating